

On the Non-Quantizability of Gravity and Constraint-Induced Rigidity of Physical Structure

A Closure Theorem via Admissibility and Divergence Exhaustion

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Abstract

This paper proves a closure theorem for primitive attempts to quantize gravity. Starting from a standard mathematical-physics representation of a quantization attempt, the argument shows that any same-domain map from GR-type constrained Hamiltonian structure to QM-type Hilbert structure must diverge in one of finitely many technical loci: representation, observable assignment, constraint closure, physical state selection, measure/inner-product choice, boundary extraction, or continuation. These loci exhaust the admissibility-relevant degrees of freedom of primitive quantization. Using the AASC admissibility framework, each divergence is then mapped to a finite normal-form failure class. Every class either imports auxiliary structure, changes scope, repairs failure post hoc, moves the target, or collapses extensionally to an already constraint-native realization. Hence gravity is not quantizable as a primitive same-domain operation. The positive result is constraint-induced rigidity: admissibility-preserving physical structure must be built as a constraint-native realization whose GR and QM presentations are bookkeeping projections of a common quotient architecture, rather than by quantizing one projection into the other. The theorem does not deny effective quantization methods; it denies their standing as primitive foundations for quantum gravity.

Roadmap

The paper is not a new dynamical model of quantum gravity. It is a closure theorem about the well-typedness of the operation normally called quantization of gravity. Section 1 fixes the AASC/AER background, the projection language, and a formal type discipline for same-domain maps. Section 2 defines primitive quantization in external mathematical-physics data, proves the quantization divergence lemma, and maps technical divergence loci into AASC normal forms. Section 3 proves minimality of the constraint-native interface with one concrete witness class for every clause. Section 4 replaces formalism-to-formalism unification by projection equivalence. Section 5 states the realization-certificate program. Sections 6–7 classify existing approaches and give failure witnesses. Section 8 gives a formal certificate for constraint-native quantum realization.

Core thesis. Gravity is not quantized as a primitive object. GR and QM are bookkeeping projections of a common admissibility-constrained quotient architecture. A successful quantum-gravity theory is therefore not a map that applies quantum rules to classical gravity, but a realization of the underlying architecture whose projections reproduce the GR and QM theorem objects in their proper roles.

1 Preliminaries: AASC, AER, projection, and type

1.1 Published dependencies

The paper treats the following as previously established AASC results.

(D1) **AASC kernel.** The kernel of admissibility-bearing construction is

$$K = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}),$$

where admissibility distinguishes licensed continuation from inadmissible construction, standing is reusable act-status, reference fixes same-domain target identity, and irreversibility blocks same-act post-hoc repair. No faithful same-domain lower generator or kernel-free countermodel exists for the target of admissibility-assessable, identity-preserving construction.

(D2) **Admissibility-interface structure.** On a fixed domain, the pre-admissible boundary is AMetric, the evaluated interface is bivalent, standing is reuse-stable admissibility, the admissible interior is unique up to identity-preserving equivalence, and standing is conserved rather than created, transferred, amplified, or repaired.

(D3) **AER Hamiltonian rigidity.** For regular-stratum observable Hamiltonian evaluation, the AER interface–extension-independent observable Poisson descent, quotient separation, and observable completeness–forces coisotropic standing, characteristic redescription, local first-class generators, and multiplier-type Hamiltonian presentation.

(D4) **Quantum interior/boundary rigidity.** Under the AASC predictive regime, admissibility-preserving predictive geometry is Hilbertian and reversible admissible dynamics takes Schroedinger/unitary form. Bell witness structure and measurement-record use are fixed-domain boundary quotient classifications.

The present paper does not re-prove these results. It applies them to the specific status of primitive GR quantization.

1.2 The common quotient architecture

Definition 1.1 (Admissibility-bearing quotient architecture). *An admissibility-bearing quotient architecture is a role-labeled object*

$$\mathcal{A} = (\text{Anc}_{\mathcal{A}}, \text{Ten}_{\mathcal{A}}, \text{Skin}_{\mathcal{A}}, \sim_{\mathcal{A}}, \mathcal{C}_{\mathcal{A}}),$$

where $\text{Anc}_{\mathcal{A}}$ are domain-fixing anchors, $\text{Ten}_{\mathcal{A}}$ are standing-bearing quotient/tensor witnesses, $\text{Skin}_{\mathcal{A}}$ is representational surplus, $\sim_{\mathcal{A}}$ is lawful identity-preserving equivalence, and $\mathcal{C}_{\mathcal{A}}$ is the class of admissible continuations. A presentation realizes $\mathcal{A}$ when its standing-bearing content factors through these data up to lawful role-preserving equivalence.

Definition 1.2 (Projection). *A projection of $\mathcal{A}$ is a role-preserving map*

$$\Pi_X : \mathcal{A} \longrightarrow X$$

whose image is a theorem object X and whose kernel consists only of skin-level redundancy relative to the fixed comparison class. Projection is not a functor from one physical formalism to another; it is a quotient-realization map from the underlying architecture to a role-specific bookkeeping presentation.

Definition 1.3 (GR and QM projections). *A GR-type projection $\Pi_{\text{GR}} : \mathcal{A} \rightarrow \mathfrak{G}$ has as image a constrained Hamiltonian theorem object: standing surface, quotient of physical observables, characteristic redescription, first-class constraint span, and multiplier-type presentation. A QM-type projection $\Pi_{\text{QM}} : \mathcal{A} \rightarrow \mathfrak{Q}$ has as image a quantum theorem object with a Hilbert/Schroedinger interior and Bell/measurement-record boundary quotients.*

Remark 1.4 (What projection does not mean). Projection does not assert that Lorentzian geometry, ADM phase space, Hilbert space, operator algebras, and measurement records are literally the same mathematical object. It asserts that their admissibility-bearing content is obtained by role-preserving quotient realization from the same underlying architecture. No ontological merger, functorial equivalence, or categorical equivalence between GR and QM formalisms is asserted.

1.3 Formal type discipline for same-domain maps

Definition 1.5 (Admissibility type of a physical construction). *The admissibility type of a physical construction or presentation X is the quadruple*

$$\text{Typ}(X) := (\text{Anc}_X, \text{Ten}_X, \sim_X, \mathcal{C}_X),$$

where Anc_X fixes domain, boundary, regime, and admissibility envelope; Ten_X is the standing-bearing tensor/witness package; $\sim_X$ is lawful identity-preserving equivalence; and $\mathcal{C}_X$ is admissible continuation/composition.

Definition 1.6 (Same-domain well-typed map). *A map $F : X \rightarrow Y$ is same-domain well typed when it preserves admissibility type up to lawful role-preserving equivalence:*

$$F_*(\text{Anc}_X) = \text{Anc}_Y, \quad F_*(\text{Ten}_X) = \text{Ten}_Y, \quad x \sim_X x' \Rightarrow F(x) \sim_Y F(x'), \quad F_*(\mathcal{C}_X) = \mathcal{C}_Y.$$

If one of these clauses fails, F is not a same-domain map; it is anchor motion, tensor-witness change, quotient motion, or continuation motion. If all clauses hold, any remaining difference is skin relative to the common quotient architecture.

Proposition 1.7 (Primitive projection-to-projection maps are type-constrained). *Let $\mathfrak{G}$ and $\mathfrak{Q}$ be GR-type and QM-type projections of the same architecture $\mathcal{A}$. Any map $F : \mathfrak{G} \rightarrow \mathfrak{Q}$ carrying standing-bearing force must either preserve admissibility type and hence factor through $\mathcal{A}$ as a relation between projections, or fail one component of same-domain type preservation. In the first case it is not primitive quantization; in the second it is a typed domain change, hidden load, or standing failure.*

Proof. The GR and QM presentations are role-specific images of $\mathcal{A}$. If F preserves anchors, tensor witnesses, quotient identity, and continuation, then F is constant on the skin-fibres of both projections and factors through the common quotient architecture. If it does not preserve one of those four components, it has changed the admissibility type of the construction. The AASC role decomposition leaves no fifth same-domain component on which independent standing-bearing force could reside. $\square$

2 Normal-form exhaustion for primitive quantization attempts

2.1 Primitive GR quantization as a same-domain claim

Standard “quantizing gravity” language presupposes a map of the form

$$\mathcal{Q}_{\text{std}} : \mathfrak{G} \longrightarrow \mathfrak{Q},$$

where $\mathfrak{G}$ is treated as a primitive classical gravitational structure and $\mathfrak{Q}$ as a separate primitive quantum target. Canonical and path-integral programs differ in implementation, but the structural posture is shared: begin with a GR-side presentation, impose or construct a quantum-side presentation, and seek consistency.

Under the AASC/AER projection architecture, this posture is not fundamental. The GR-side object $\mathfrak{G}$ and the QM-side object $\mathfrak{Q}$ are already projections of $\mathcal{A}$. A map $\mathfrak{G} \rightarrow \mathfrak{Q}$ is therefore either a relation between projections or a disguised construction of a new realization of $\mathcal{A}$. It cannot be a primitive same-domain operation unless it preserves anchors, tensors, quotient identity, and admissible continuation.

2.2 External quantization data

Definition 2.1 (External primitive quantization tuple). *Before applying AASC terminology, a primitive quantization attempt for gravity is an externally specified mathematical-physics tuple*

$$\mathbf{Q} = (P_{\text{cl}}, \theta_{\text{cl}}, C_a, \mathcal{O}_{\text{cl}}, \rho, \mathcal{H}, \mathcal{D}, \hat{\rho}, \hat{C}_a, \langle \cdot, \cdot \rangle_{\text{phys}}, \mu, \text{Obs}_{\text{out}}). \quad (2.1)$$

Here:

- (Q1) $(P_{\text{cl}}, \theta_{\text{cl}}, C_a)$ is a classical constrained phase-space presentation, where θ_{cl} is Poisson/symplectic data and $C_a \approx 0$ are classical constraints or gauge generators;
- (Q2) $\mathcal{O}_{\text{cl}}$ is the classical observable algebra or selected classical quantities to be quantized;
- (Q3) $\rho : \mathcal{O}_{\text{cl}} \rightarrow \text{Op}_{\text{formal}}$ is an assignment of classical quantities to formal operators;
- (Q4) $\mathcal{H}$ is a candidate kinematical state space;
- (Q5) $\mathcal{D}$ specifies operator domains, dense subspaces, rigging maps, distributional extensions, or physical-state domains;
- (Q6) $\hat{\rho}$ is the realized operator representation of the observable and constraint data on $\mathcal{D} \subseteq \mathcal{H}$;
- (Q7) $\hat{C}_a$ are quantum constraints and their closure/anomaly data;
- (Q8) $\langle \cdot, \cdot \rangle_{\text{phys}}$ is the physical inner product, reduced inner product, group-averaged form, or equivalent state-comparison datum;
- (Q9) μ denotes measure, regularization, contour, gauge-fixing, boundary, graph/refinement, or comparable construction data whenever the tuple is path-integral, covariant, lattice, loop, or string-like rather than purely canonical;
- (Q10) Obs_{out} specifies how amplitudes, observables, records, or predictions are extracted.

The tuple is called *primitive* when it is claimed to take the GR-side object as input and produce the quantum object without first constructing a common admissibility-bearing architecture whose projections are GR-like and QM-like.

Definition 2.2 (AASC primitive quantization attempt). *An AASC primitive quantization attempt is an external tuple $\mathbf{Q}$ together with the claim that its choices determine a same-domain standing-bearing map*

$$Q : \mathfrak{G} \rightarrow \mathfrak{Q}$$

that preserves the fixed admissibility-bearing target while carrying foundational force not already fixed by the common architecture $\mathcal{A}$.

2.3 Technical divergence before AASC translation

Definition 2.3 (Technical coordinates of a primitive quantization tuple). *Let $\mathbf{Q}$ be the external primitive quantization tuple of (2.1). Its standing-bearing technical coordinates are the ordered list*

$$\text{Tech}(\mathbf{Q}) = (L_1(\mathbf{Q}), \dots, L_7(\mathbf{Q})),$$

where:

- (L1) **State-space locus:** $\mathcal{H}$, representation, graph space, Fock space, string state space, categorical state object, or equivalent state carrier;
- (L2) **Observable-map locus:** ρ and $\hat{\rho}$, including factor ordering, regularization of composite operators, and representation of classical observables;
- (L3) **Constraint-closure locus:** $\hat{C}_a$, their domains, anomaly terms, algebraic closure, BRST/BV structure, reduced/Dirac constraint implementation, and closure certificates;
- (L4) **Physical-domain and inner-product locus:** $\mathcal{D}$, physical state space, inner product, rigging map, group averaging, superselection sector, or state-comparison structure;
- (L5) **Measure/regularization/boundary locus:** path-integral measure, gauge fixing, contour, regulator, triangulation, spin-foam sum, boundary condition, compactification, asymptotic data, or UV datum;
- (L6) **Continuation locus:** time/evolution rule, history composition, refinement/continuum limit, unitary continuation, admissible record lineage, or composition law for physical histories;
- (L7) **Output/record locus:** extraction of observables, amplitudes, S -matrix elements, semiclassical limits, measurement records, Bell/record data, or predictive outputs.

For two tuples $\mathbf{Q}$ and $\mathbf{Q}'$ with the same declared classical input and target comparison class, write $\mathbf{Q} \equiv_i \mathbf{Q}'$ when their i -th technical coordinates agree up to conventional reparameterization that preserves all already fixed earlier coordinates. Write $\mathbf{Q} \equiv_{1\dots i} \mathbf{Q}'$ when $\mathbf{Q} \equiv_j \mathbf{Q}'$ for all $j \leq i$.

Definition 2.4 (First standing-bearing technical divergence). *A first standing-bearing technical divergence between $\mathbf{Q}$ and $\mathbf{Q}'$ occurs at locus L_i when:*

- (D1) $\mathbf{Q} \equiv_{1\dots i-1} \mathbf{Q}'$, with the condition vacuous for $i = 1$;
- (D2) $\mathbf{Q} \not\equiv_i \mathbf{Q}'$; and
- (D3) the difference at L_i changes at least one component of the admissibility type of the resulting construction—anchor, tensor witness, quotient identity, continuation, or output/record standing—rather than merely changing notation.

The ordering of loci is an audit ordering, not a claim of temporal priority. It provides a finite normal-form coordinate system for the first place where a primitive quantization attempt can acquire standing-bearing force.

Theorem 2.5 (Quantization divergence lemma, formal form). *Let $\mathbf{Q}$ and $\mathbf{Q}'$ be external primitive quantization tuples for the same declared classical input and the same claimed quantum-gravity target. If they are not skin-equivalent as same-domain quantization attempts, then there exists $i \in \{1, \dots, 7\}$ such that $\mathbf{Q}$ and $\mathbf{Q}'$ have a first standing-bearing technical divergence at locus L_i . Conversely, if $\mathbf{Q} \equiv_{1\dots 7} \mathbf{Q}'$, then no standing-bearing quantization divergence remains: the two tuples determine the same admissibility type and differ only by skin relative to the fixed target.*

Proof. The external tuple (2.1) exposes every datum by which a primitive quantization attempt can affect same-domain standing: the carrier of states (L1), the assignment of classical quantities to quantum representatives (L2), the closure and anomaly behavior of constraints (L3), the physical domain and state-comparison structure (L4), measure/regularization/boundary data (L5), admissible continuation or history composition (L6), and extraction of predictions or records (L7). These seven coordinates exhaust the admissibility-relevant degrees of freedom of the external tuple.

If two tuples agree at all seven loci up to the allowed reparameterizations, then they have the same state candidates, represented observables, constraint algebra and anomaly data, physical domains and inner products, regularized or measured amplitudes, admissible continuation structure, and output/record extraction. They therefore have the same anchors, tensor witnesses, quotient relation, continuation, and output-standing profile. Any remaining difference is a change of presentation and is skin.

If the tuples are not skin-equivalent, then at least one admissibility-type component changes. Because the list of loci is finite, choose the least index i at which such a standing-bearing difference appears after quotienting out agreement at earlier loci. By construction this is a first standing-bearing technical divergence at L_i . Hence every genuine primitive quantization divergence has a location in the finite list before any AASC normal-form translation is applied. $\square$

2.4 AASC normal forms

Definition 2.6 (Primitive quantization normal-form family). *A proposed primitive quantization attempt has an AASC normal-form image in the following finite family:*

- (N1) **Skin factorization:** *the map changes only presentation and factors through $\mathcal{A}$.*
- (N2) **Anchor motion:** *the map changes domain, background, boundary data, regulator envelope, or admissibility conditions.*
- (N3) **Tensor-witness change:** *the map adds, deletes, or changes physical observables, predictive witnesses, anomaly data, records, or invariant amplitudes.*
- (N4) **Quotient motion:** *the map changes the equivalence relation identifying same physical content.*
- (N5) **Continuation motion:** *the map changes admissible histories, constraint closure, composition, or physical evolution.*
- (N6) **Selector import:** *the map uses an ordering, measure, preferred representation, canonical representative, factor ordering, branch choice, or contour with standing-bearing force.*
- (N7) **Post-selection repair:** *the map generates inadmissible candidates and filters or repairs them while treating the result as the same construction.*
- (N8) **Boundary transport:** *the map uses interior quantum or metric structure to decide pre-boundary admissibility or record fixation.*
- (N9) **Target rejection:** *the map declines the fixed-domain target and constructs a different theorem object.*

Lemma 2.7 (Technical loci map to AASC normal forms). *Each technical divergence locus (L1)–(L7) has a nonempty AASC normal-form image in (N1)–(N9):*

- (M1) *L1 state space: if declared, anchor or tensor witness; if hidden, selector import or quotient motion.*
- (M2) *L2 observable map: if declared, tensor witness; if hidden, selector import, or skin if inert.*

- (M3) *L3 constraint closure: if declared, continuation or tensor witness; if hidden, standing failure or post-selection repair.*
- (M4) *L4 physical domain / inner product: if declared, anchor or quotient structure; if hidden, selector import or quotient motion.*
- (M5) *L5 measure / regularization / boundary: if declared, anchor or tensor data; if hidden, boundary transport or selector import.*
- (M6) *L6 continuation: if declared, continuation structure; if hidden, standing failure or target rejection.*
- (M7) *L7 output / record extraction: if declared, tensor or boundary quotient; if hidden, hidden load or post-selection repair.*

Proof. A declared divergence is part of the theorem object and appears as anchor, tensor, quotient, or continuation motion. An undeclared divergence carrying standing-bearing force is selector import, hidden load, boundary transport, or repair, depending on where it enters. A divergence with no standing-bearing effect is skin factorization. The table enumerates exactly the seven external loci of Lemma 2.5. No additional AASC normal form is available because the AASC role decomposition contains no fourth same-scope role beyond anchors, tensors, and skin, and the fixed-domain comparison class contains no additional locus beyond quotient and continuation. $\square$

Lemma 2.8 (Normal-form image is nonempty). *Every genuine primitive quantization attempt has nonempty AASC normal-form image in the family (N1)–(N9).*

Proof. Take its first non-skin technical divergence. By Lemma 2.5, the divergence lies in (L1)–(L7). By the technical-locus-to-normal-form lemma, that locus maps to at least one of (N1)–(N9). Hence the AASC normal-form image is nonempty. $\square$

Theorem 2.9 (Collapse of the primitive quantization normal forms). *No normal form (N1)–(N9) survives as an independent same-domain primitive quantization route.*

Proof. (N1) is not primitive quantization; it is factorization through $\mathcal{A}$. (N2) changes scope. (N3) changes tensor content and therefore the theorem object. (N4) changes identity of physical content. (N5) changes lawful continuation or dynamics. (N6) is an inadmissible second same-domain classifier unless it is declared as new tensor or anchor load. (N7) violates AASC irreversibility and no same-act repair. (N8) violates the AMetric boundary and boundary non-transmissivity: post-admission structure cannot govern pre-admission classification. (N9) is a legitimate change of target, not a same-domain quantization route. Thus every normal form either collapses to bookkeeping, changes domain, imports explicit new load, or fails standing. $\square$

Theorem 2.10 (Primitive Quantization Obstruction). *Let $\mathcal{A}$ be a fixed AASC/AER admissibility-bearing quotient architecture with GR and QM projections $\Pi_{GR} : \mathcal{A} \rightarrow \mathfrak{G}$ and $\Pi_{QM} : \mathcal{A} \rightarrow \mathfrak{Q}$. A primitive same-domain quantization map $Q : \mathfrak{G} \rightarrow \mathfrak{Q}$ carrying independent foundational force does not exist. Any such map either factors through $\mathcal{A}$, changes scope, imports auxiliary standing-bearing load, or fails standing.*

Proof. By the nonempty normal-form theorem, Q has a nonempty image in (N1)–(N9). By the normal-form collapse theorem, no such image survives as independent same-domain primitive quantization. Hence a primitive same-domain quantization map with independent foundational force does not exist. $\square$

Corollary 2.11 (No primitive $\mathfrak{G} \rightarrow \mathfrak{Q}$ foundation). *A GR-to-QM map cannot be the foundational relation between gravity and quantum structure on the fixed AASC/AER domain. If it is admissibility-preserving, it is a derived bookkeeping relation between projections; if it is not, it changes scope, imports load, or fails standing.*

3 Minimality of the constraint-native obstruction

The previous theorem rules out primitive quantization attempts under the full AASC/AER comparison structure. A hostile response might ask whether the interface is too strong. This section answers that objection directly: weakening any load-bearing clause leaves room for non-equivalent same-scope realizations or untyped quantization choices.

Definition 3.1 (Constraint-native quantum-gravity interface). *The constraint-native quantum-gravity interface QGI consists of the following clauses for a candidate realization:*

- (I1) **Anchor fixation:** *domain, boundary data, scope, error envelope, and admissibility envelope are fixed.*
- (I2) **Tensor-witness completeness:** *all standing-bearing observables, amplitudes, predictive witnesses, anomaly data, and record witnesses are represented.*
- (I3) **Quotient identity:** *lawful identity-preserving equivalence is fixed and preserved by presentation changes.*
- (I4) **Continuation closure:** *admissible histories, constraint algebra, composition, and reversible predictive continuation are fixed.*
- (I5) **Observable completeness:** *no nonzero same-scope physical direction or quantum choice remains invisible to the witness algebra while carrying standing-bearing force.*
- (I6) **Boundary non-transport:** *interior metrics, operators, measures, and selectors do not govern pre-admissible boundary or record fixation unless explicitly reified as state-level structure.*

Definition 3.2 (Strict weakening). *A strict weakening of QGI is an interface obtained by omitting or weakening at least one of clauses (I1)–(I6), while still claiming to classify primitive quantization attempts on the same fixed target.*

Theorem 3.3 (QGI weakening witness ledger). *For each load-bearing clause of QGI, omitting or weakening that clause opens a concrete non-equivalent realization, untyped quantization choice, or standing-failure mode as recorded in the following theorem-table.*

<i>Weakened clause</i>		<i>Failure mode reopened</i>	<i>Concrete witness class</i>
<i>Anchor fixation</i>		<i>Anchor motion / scope drift</i>	<i>Regulator, background, boundary condition, compactification, graph/refinement class, or asymptotic state choice changes amplitudes or observables while still being treated as the same problem.</i>
<i>Tensor-witness completeness</i>	<i>com-</i>	<i>Hidden tensor load</i>	<i>Anomaly term, counterterm, composite-operator renormalization, stress-energy witness, Bell/record datum, or predictive amplitude affects outputs without appearing in the theorem object.</i>
<i>Quotient identity</i>		<i>Quotient motion / representative privilege</i>	<i>Gauge, diffeomorphism, foliation, basis, graph embedding, or representative choice becomes physical by presentation rather than invariant content.</i>
<i>Continuation closure</i>		<i>Continuation failure / standing failure</i>	<i>Quantum constraints fail to close, histories fail to compose, BRST/BV differential fails, group averaging fails to preserve domain, or evolution is not well defined on the quotient.</i>
<i>Observable completeness</i>	<i>complete-</i>	<i>Hidden same-scope remainder</i>	<i>The incomplete presymplectic quotient of Section 7.1: a nonzero same-scope direction is invisible to all physical observables.</i>
<i>Boundary transport</i>	<i>non-</i>	<i>Boundary selector / illicit transport</i>	<i>Path-integral measure, branch selector, typicality measure, canonical representative, or metric distance defined only after admission is used to decide boundary or record fixation.</i>

Thus each QGI clause has an explicit witness showing why the clause is needed to close primitive quantization as a same-domain route.

Proof. Each row gives a constructional witness with the same form: when the named clause is removed, a datum that was required to be fixed, represented, quotiented, continued, observed, or kept boundary-inert can acquire standing-bearing force while remaining outside the declared same-domain architecture. The failure mode column names the corresponding AASC normal form. Since every row changes an anchor, tensor witness, quotient relation, continuation, observable-completeness condition, or boundary-status condition, each row supplies an actual reopened route rather than a merely verbal possibility. $\square$

Theorem 3.4 (Minimality of the constraint-native interface). *Every strict weakening of QGI admits a non-equivalent same-scope realization, an untyped quantization choice, or a standing-failure mode. Hence the Primitive Quantization Obstruction is minimal with respect to admissibility-bearing interface clauses: no weaker interface rules out primitive quantization attempts while preserving the same target.*

Proof. Use the QGI weakening witness ledger, Theorem 3.3. If anchor fixation is weakened, the regulator, boundary, background, compactification, asymptotic condition, or graph/refinement choice may alter amplitudes or observables while the construction still claims same-domain status.

This is anchor motion rather than a same-scope quantization. If tensor-witness completeness is weakened, anomaly terms, counterterms, predictive amplitudes, records, or stress-energy witnesses may affect standing-bearing outputs while remaining absent from the theorem object; this is hidden tensor load. If quotient identity is weakened, gauge, diffeomorphism, foliation, basis, graph embedding, or representative choices can become physical by presentation; this is quotient motion or representational privilege. If continuation closure is weakened, the quantum constraints, BRST/BV differential, history composition, refinement limit, or predictive evolution can fail to be well defined on the quotient; this is continuation motion or standing failure. If observable completeness is weakened, the concrete presymplectic model of Section 7.1 exhibits a closed quotient with a nonzero same-scope remainder invisible to all physical observables. If boundary non-transport is weakened, post-admission structures such as measures, contours, canonical representatives, or branch selectors can decide pre-boundary or record-level standing; this violates the AMetric boundary or imports a selector. Thus every strict weakening reopens at least one normal form (N2)–(N8). Therefore the interface is minimal for excluding primitive quantization while preserving the fixed target. $\square$

Corollary 3.5 (No weaker quantization criterion). *Any criterion weaker than QGI that still permits primitive quantization of GR to claim same-domain standing leaves at least one normal-form route open. Therefore a successful quantum-gravity construction must either satisfy QGI or explicitly declare a richer scope.*

4 Projection equivalence instead of formalism-to-formalism unification

Definition 4.1 (Projection equivalence). *Two theorem objects X and Y are projection-equivalent when there exists a common admissibility-bearing quotient architecture $\mathcal{A}$ and role-preserving projections $\Pi_X : \mathcal{A} \rightarrow X$, $\Pi_Y : \mathcal{A} \rightarrow Y$ such that all standing-bearing differences in X and Y are images of anchor/tensor differences in $\mathcal{A}$, and all remaining differences are skin in their respective presentations.*

Theorem 4.2 (Projection Equivalence of GR and QM theorem objects). *Within the AASC/AER comparison class, the GR theorem object $\mathfrak{G}$ and the QM theorem object $\mathfrak{Q}$ are projection-equivalent. The equivalence is at the level of admissibility-bearing quotient architecture and not at the level of direct formal conversion between GR and QM mathematical presentations.*

Proof. The GR projection has as standing-bearing content the constrained Hamiltonian interior: observable quotient, characteristic redescription, first-class closure, and multiplier presentation. The QM projection has as standing-bearing content the Hilbert/Schroedinger predictive interior together with Bell witness and measurement-record boundary quotients. By AASC/AER rigidity, each theorem object factors through the anchor/tensor/skin architecture with no second same-scope classifier and no hidden same-scope remainder. Hence there exists a common quotient architecture $\mathcal{A}$ from which both objects are obtained by role-preserving projection. Since the formal categories of symplectic geometry and Hilbert/operator structure are different presentations of different roles, no direct functorial or ontological identity is implied. $\square$

Corollary 4.3 (Quantization replaced by realization). *The fundamental problem is not to construct a map $\mathfrak{G} \rightarrow \mathfrak{Q}$. It is to construct a realization $\mathcal{R}$ of $\mathcal{A}$ whose projections recover $\mathfrak{G}$ and $\mathfrak{Q}$ in their respective roles.*

5 The positive program: realization certification

Definition 5.1 (Constraint-native realization). *A constraint-native realization is a structure*

$$\mathcal{R} = (\mathcal{A}_{\mathcal{R}}, \Pi_{\text{GR}}^{\mathcal{R}}, \Pi_{\text{QM}}^{\mathcal{R}}, \mathcal{E}_{\mathcal{R}})$$

where $\mathcal{A}_{\mathcal{R}}$ realizes the AASC/AER quotient architecture, $\Pi_{\text{GR}}^{\mathcal{R}}$ produces a GR-type constrained Hamiltonian interior, $\Pi_{\text{QM}}^{\mathcal{R}}$ produces a QM-type Hilbert/Schroedinger interior plus boundary witness/record structure, and $\mathcal{E}_{\mathcal{R}}$ is an explicit admissibility envelope specifying anchors, tensor witnesses, quotient identity, and continuation.

Definition 5.2 (Constraint-native quantum gravity problem). *The constraint-native quantum gravity problem is to construct $\mathcal{R}$ satisfying:*

- (QG1) AASC kernel and admissibility-interface conditions;
- (QG2) AER- $\text{Obs}_{\min}$, AER-Sep, and AER-Max for the Hamiltonian interior;
- (QG3) anomaly-free or closure-preserving continuation of the constraint/redescription algebra;
- (QG4) Hilbert/Schroedinger predictive interior on the quantum projection;
- (QG5) Bell/measurement-record boundary quotient well-definedness;
- (QG6) agreement with the GR and QM theorem objects in the relevant classical/quantum regimes.

Theorem 5.3 (Realization-certificate theorem). *Any candidate supplying the data (QG1)–(QG6) is not a primitive quantization of gravity but a joint GR/QM constraint-native realization. Conversely, any putative quantum-gravity theory that claims to preserve both GR and QM standing-bearing content must supply these data explicitly or be classified as skin, scope change, auxiliary load, or standing failure.*

Proof. Conditions (QG1)–(QG2) ensure the GR interior has constrained Hamiltonian rigidity. Conditions (QG3)–(QG4) ensure that the quantum interior is admissibility-preserving and Hilbert/Schroedinger in the predictive regime. Condition (QG5) ensures that boundary witness and record structures are well-defined. Condition (QG6) supplies physical matching to the usual theorem objects. For the converse, a theory preserving both GR and QM standing-bearing content must fix the same anchors, tensor witnesses, quotient identities, and continuation structures, or else move one of them. The AASC role decomposition then classifies the theory exactly as stated. $\square$

Corollary 5.4 (No metric-first or Hilbert-first primacy). *Neither metric geometry nor Hilbert/operator structure is primitive in the constraint-native problem. Each is a projectional bookkeeping form of $\mathcal{A}$, and either may fail to be fundamental in a deeper realization while remaining fully valid in its projected regime.*

6 Classification of existing quantum-gravity strategies

The classifications in this section are diagnostic sketches, not full evaluations of the technical programs. Their purpose is to show how the AASC obstruction lands on familiar quantization machinery. A complete assessment of any program would require a full realization certificate in the sense of Section 8.

Definition 6.1 (Strategy classes). *A quantum-gravity strategy is typed as one of the following relative to $\mathcal{A}$:*

- (S1) **Quotient-compatible bookkeeping:** it changes presentation but no standing-bearing content;
- (S2) **Richer-scope realization:** it adds explicit state variables, boundary data, tensor witnesses, or continuation rules;
- (S3) **Auxiliary-load import:** it uses selectors, measures, representations, regulator choices, or orderings with standing-bearing force without declaring them as part of the theorem object;
- (S4) **Standing failure:** it fails to preserve quotient identity, closure, or admissible continuation.

Proposition 6.2 (Canonical and Dirac quantization diagnostic). *Canonical quantization of ADM-type data is well typed only after it supplies the external tuple data of Definition 2.3: an operator representation of constraints, domains, physical state space, inner product, anomaly/closure control, and observable extraction. If factor ordering, representation choice, dense domain, rigging map, constraint regularization, or physical inner product has no standing-bearing effect, it is skin. If it has standing-bearing effect and is declared, the construction is a richer-scope realization. If it is not declared, it is auxiliary-load import. If the quantum constraint algebra fails to close or is not represented on the physical domain, the construction fails standing.*

Proof. Dirac quantization requires a representation of classical constraints by operators and a physical state space solving those constraints. Factor ordering changes the observable-map locus; domain and rigging choices change the physical-domain locus; anomaly control changes the constraint-closure locus; physical observables and records change the output locus. By Lemma 2.5, any first standing-bearing divergence appears at one of these loci. The AASC normal-form map then classifies it as skin, declared richer-scope structure, hidden load, or standing failure. $\square$

Proposition 6.3 (Path-integral and covariant quantization diagnostic). *A gravitational path integral is same-domain well typed only if its integration domain, gauge quotient, Faddeev-Popov/BRST or BV data, measure, contour, boundary conditions, regularization, and observable extraction are typed in the anchor/tensor/skin ledger. If those data are inert, they are skin. If they affect amplitudes, records, or observables, they are anchors or tensor witnesses. If they affect outputs without being declared, they are auxiliary load; if gauge fixing or quotienting fails, standing fails.*

Proof. The path-integral expression is not an output until a measure, domain, gauge quotient, boundary condition, and extraction map are fixed. These are technical loci (L5), (L3), (L4), and (L7). A choice that changes no quotient content is bookkeeping. A choice that changes amplitudes or observables is standing-bearing and must be part of the theorem object. An undeclared standing-bearing choice is hidden load. Failure of BRST/BV or gauge quotient closure is failure of continuation and quotient identity. $\square$

Proposition 6.4 (Loop, spin-network, and spin-foam diagnostic). *Loop, spin-network, and spin-foam variables can be quotient-compatible skins or richer-scope realizations, but not primitive same-domain quantization maps unless graph/refinement data, diffeomorphism quotient, Hamiltonian or master constraint, continuum limit, and physical inner product are all typed in the QGI ledger. Graph discreteness, embedding data, refinement rules, or foam amplitudes with standing-bearing force are tensor or anchor data, not mere representation.*

Proof. The state-space locus is occupied by loop or spin-network state data; quotient identity is controlled by gauge and diffeomorphism equivalence; continuation is controlled by the Hamiltonian/master constraint or foam transition amplitudes; the physical inner product and continuum/refinement map occupy loci (L4)–(L6). If these choices preserve the same AASC quotient,

they are skin or lawful projection data. If they change standing-bearing content, they are richer-scope structure. If they are used without declaration, they are auxiliary load. A failure of constraint/refinement closure is standing failure. $\square$

Proposition 6.5 (String-like and UV-completion diagnostic). *String-like or other UV-completion programs are most naturally typed as richer-scope realizations rather than primitive quantizations of GR. If successful, they supply a candidate $\mathcal{R}$ whose low-energy projection is GR-like and whose quantum projection is Hilbert/operator-like. Their extra dimensions, worldsheet data, brane sectors, dualities, compactification data, and moduli are anchors or tensor witnesses when they affect standing-bearing outputs.*

Proof. A UV completion introduces structure outside the original GR projection. When the added structure is declared and controls low-energy projection, anomaly cancellation, spectra, amplitudes, or boundary data, it is a richer-scope realization. If a duality changes only presentation, it is skin relative to the quotient architecture. If compactification, moduli choice, or background data affect outputs without being included in the theorem object, the construction imports hidden load. Thus the program may be admissible, but it is not primitive quantization of GR. $\square$

Remark 6.6 (Scope of the diagnostics). The preceding propositions do not claim that canonical, path-integral, loop, spin-foam, string-like, or EFT programs fail. They identify the certificate burden each program must meet to count as constraint-native rather than primitive projection-to-projection quantization. A successful program should be read as supplying a realization certificate, not as refuting the obstruction theorem.

7 Failure witnesses and acceptance criteria

7.1 Incomplete presymplectic quotient witness

Consider

$$S = \mathbb{R}^4_{(q_1, p_1, q_2, p_2)}, \quad \omega = dq_1 \wedge dp_1 + dq_2 \wedge dp_2,$$

and let

$$C = \{p_1 = 0\}.$$

Then C is coisotropic and

$$\ker(\omega_C) = \text{span}\{\partial_{q_1}\}.$$

Define an incomplete quotient map

$$\pi : C \rightarrow \mathbb{R}, \quad \pi(q_1, q_2, p_2) = q_2.$$

Then

$$D = \ker(d\pi) = \text{span}\{\partial_{q_1}, \partial_{p_2}\},$$

so $\ker(\omega_C) \subsetneq D$. Physical observables are functions of q_2 , so AER-Obs_{min} and AER-Sep hold, but AER-Max fails: $\partial_{p_2} \notin \ker(\omega_C)$ and no physical observable detects it.

Proposition 7.1 (Witness of weakening AER-Max). *The above model is a closed but observable-incomplete quotient. It exhibits a same-scope redescription distribution strictly larger than the characteristic foliation. Therefore any interface lacking observable completeness admits a non-equivalent realization of the quotient data and cannot force constrained Hamiltonian rigidity.*

Proof. All observables factor through q_2 , and their Poisson brackets descend trivially. Separation relative to the declared quotient holds because dq_2 spans the annihilator of D . Yet D contains ∂_{p_2} , which is nonzero in the presymplectic quotient and invisible to all physical observables. Hence exact characteristic identification fails. This is a concrete realization of the minimality theorem for observable completeness. $\square$

7.2 Anchor-fixation witness: boundary data

A gravitational path integral or canonical transition amplitude with different asymptotic boundary data, topology class, compactification, regulator envelope, or graph/refinement class can yield different amplitudes while being written with the same symbolic action. If those data affect outputs, they are anchors of the theorem object. Treating them as mere convention while using them to fix amplitudes is anchor motion.

7.3 Tensor-witness witness: anomaly and counterterm data

A quantum constraint algebra with a regulator-dependent anomaly term or counterterm may change the physical state space or observable algebra. If the term has no physical effect, it is skin. If it affects constraints, amplitudes, or records, it is a tensor witness. Omitting it from the theorem object while relying on its effect is hidden tensor load.

7.4 Quotient-identity witness: representative-dependent physicality

If a gauge, diffeomorphism, foliation, graph-embedding, or basis representative changes physical predictions, then the representative has become standing-bearing. Either the quotient relation was incomplete, or the representative choice is additional state-level data. It cannot remain a same-domain convention.

7.5 Continuation-closure witness: anomalous constraints

If quantum constraints $\hat{C}_a$ fail to close on the physical domain, or if their commutator generates terms not represented in the declared tensor ledger, then admissible continuation is not well defined. Closure failure is therefore not a technical nuisance but a standing failure at the continuation locus.

7.6 Concrete witness: operator ordering as hidden load

Consider a canonical scheme that assigns a classical constraint $H(q, p) = 0$ to a quantum operator $\hat{H}$. Different orderings $\hat{H}_1, \hat{H}_2$ may agree formally at leading order but differ in domains, anomaly terms, or solution spaces.

Proposition 7.2 (Ordering witness). *If $\hat{H}_1$ and $\hat{H}_2$ determine the same physical quotient, the same solution space up to admissibility-preserving equivalence, and the same record/observable content, then the ordering difference is skin. If they determine different solution spaces, observables, anomalies, or records, then the ordering choice is standing-bearing load and must be declared as part of the theorem object.*

Proof. The distinction either changes the quotient architecture or it does not. If it does not, it is skin. If it does, then it moves tensor witnesses or continuation structure. The AASC role decomposition leaves no third possibility. $\square$

7.7 Concrete witness: path-integral measure as hidden load

A gravitational path integral often has the schematic form

$$Z = \int \frac{\mathcal{D}g}{\text{Diff}} e^{iS[g]}.$$

The notation hides choices of measure, gauge fixing, boundary conditions, contour, and observable extraction.

Proposition 7.3 (Measure witness). *If two choices of measure or contour produce the same admissibility-bearing quotient content, they are bookkeeping. If they produce different amplitudes, records, or physical observables, the choice is tensor load or anchor motion and cannot be treated as a same-domain quantization convention.*

Proof. A measure or contour with no standing-bearing effect is skin. A measure or contour with standing-bearing effect changes admitted observables or continuation. Therefore it is part of the theorem object or a scope change. $\square$

7.8 Acceptance criteria for a constraint-native model

A future model offered as constraint-native quantum gravity should supply the following certificate.

- (C1) **Anchor ledger:** the domain, boundary data, regime, and admissibility envelope.
- (C2) **Tensor ledger:** invariant observables, constraint witnesses, predictive witnesses, record witnesses.
- (C3) **Skin ledger:** gauge, coordinate, basis, representation, regulator, and ordering freedoms proven inert.
- (C4) **Hamiltonian projection:** coisotropic standing, characteristic redescription, first-class closure, multiplier form or their admissible equivalent.
- (C5) **Quantum projection:** Hilbert/Schroedinger predictive interior or admissibility-equivalent structure.
- (C6) **Boundary projection:** Bell/record or measurement-output quotient well-definedness.
- (C7) **Transport certificate:** explicit maps between regimes preserving quotient identity and admissible continuation.

This checklist is not an additional physical postulate. It is the finite visibility condition for claiming that a construction realizes $\mathcal{A}$ rather than merely quantizes a projected GR skin.

8 A formal certificate for constraint-native realization

Definition 8.1 (Realization certificate). *A realization certificate for a candidate $\mathcal{R}$ consists of the following data.*

- (R1) *A specified admissibility-bearing architecture $\mathcal{A}_{\mathcal{R}}$ with anchors, tensor witnesses, quotient relation, and continuation class.*
- (R2) *A GR projection $\Pi_{\text{GR}}^{\mathcal{R}}$ exhibiting a standing surface, observable quotient, characteristic/redescription distribution, first-class closure, and multiplier presentation or an admissibility-equivalent substitute.*

- (R3) A QM projection $\Pi_{\text{QM}}^{\mathcal{R}}$ exhibiting predictive Hilbert geometry, Schroedinger/unitary reversible dynamics or an admissibility-equivalent substitute, and boundary witness/record quotients.
- (R4) A transport ledger specifying which changes of gauge, basis, regulator, cutoff, triangulation, representation, foliation, or boundary presentation are skin, and which are anchors or tensors.
- (R5) A matching ledger specifying the regimes in which the projections reproduce standard GR, QFT, and quantum-mechanical outputs within declared error envelopes.

Theorem 8.2 (Certificate sufficiency). *A candidate $\mathcal{R}$ that supplies a realization certificate satisfying (R1)–(R5) is not a primitive quantization of gravity but a constraint-native quantum realization. Its GR and QM outputs are projections of $\mathcal{A}_{\mathcal{R}}$, and any remaining presentation freedom is skin relative to the declared certificate.*

Proof. By (R1), the candidate has an admissibility-bearing architecture. By (R2), the gravitational side appears as a constrained Hamiltonian projection. By (R3), the quantum side appears as predictive and boundary quantum projections. By (R4), all representation-dependent choices are typed as anchor, tensor, or skin. By (R5), the physical regimes are matched. Hence the construction realizes the projection architecture directly. Since neither GR nor QM is used as the primitive target of the other, the construction is constraint-native. $\square$

Corollary 8.3 (Failure of certificate is diagnostic). *If a candidate cannot supply a realization certificate, the missing item identifies the failure mode: missing anchors yield undefined scope; missing tensor witnesses yield hidden load; missing quotient identity yields representation dependence; missing continuation yields failed dynamics; missing matching yields no physical projection.*

9 Implications

9.1 Why traditional quantization struggles

Traditional quantization starts with a projected object—the GR constrained Hamiltonian presentation—and attempts to impose a second projected object—the quantum Hilbert/operator presentation. But projection-to-projection maps must account for anchors, tensor witnesses, quotient identity, admissible continuation, and record extraction. The familiar difficulties of quantum gravity—constraint closure, factor ordering, measure definition, background independence, gauge redundancy, problem of time, and observable construction—are naturally reclassified as failures to construct the underlying quotient architecture rather than failures to quantize gravity as such.

9.2 Problem of time

The problem of time is not merely the absence of a preferred time variable in GR or the presence of a unitary parameter in QM. It is the mismatch between two projections: GR encodes presentation freedom and constraint-generated redescription, while QM encodes admissible reversible continuation on predictive state space. A constraint-native realization must construct a common continuation structure whose projections appear as lapse/shift arbitrariness on the GR side and Schroedinger/unitary evolution or its admissible generalization on the QM side.

9.3 Semiclassical gravity

Semiclassical gravity is naturally typed as a mixed projection regime: a GR-type geometric book-keeping skin is coupled to QM-type predictive matter witnesses. Its status depends on whether the

coupling is quotient-complete and standing-preserving. If stress-energy expectation values are used as tensor witnesses without corresponding record/observable closure, the construction is incomplete. If the coupling is explicitly effective with an error envelope, it is admissible as a restricted-scope projection.

9.4 Effective field theory of gravity

The effective field theory of gravity is not challenged by the obstruction theorem. It is a lawful restricted-scope construction when the error discipline and retained operators are part of the anchor/tensor ledger. It is not primitive quantum gravity; it is an admissible low-energy projection.

10 Non-claims and scope limits

The manuscript does not claim that no theory of quantum gravity exists. It claims that the usual operation “quantize GR” is not the primitive fixed-domain operation under the AASC/AER architecture. Nor does it claim that every standard program is invalid. It classifies standard programs by the role of their additional structure. A program that supplies explicit new anchors, tensor witnesses, boundary data, or continuation rules is a richer-scope realization. A program that preserves all standing-bearing content while changing only presentation is quotient-compatible bookkeeping. A program that relies on undeclared selectors, measures, domains, or representatives has hidden load. A program that loses quotient identity or continuation fails standing.

Proposition 10.1 (No conflict with effective practice). *Perturbative quantum gravity, semiclassical gravity, and gravitational effective field theory are compatible with the obstruction theorem when they are treated as restricted-scope projections with explicit error envelopes and retained witness ledgers.*

Proof. The obstruction theorem concerns primitive same-domain quantization. Effective constructions are not primitive same-domain maps; they specify regimes, truncations, and errors. When those are anchors in the declared scope and the retained operators are tensor witnesses, the construction is admissible as an effective projection. $\square$

11 Conclusion

The standard question “Can gravity be quantized?” assumes that GR is a primitive classical object and quantum mechanics a separate primitive target. Under the AASC/AER projection architecture, that assumption fails. GR and QM are role-specific bookkeeping projections of a common admissibility-bearing quotient architecture. Therefore primitive GR quantization is not a well-typed same-domain operation. It either factors through the architecture as bookkeeping, imports auxiliary standing-bearing load, changes scope, or fails to preserve standing.

The replacement is constructive. A quantum-gravity program should build constraint-native realizations: structures whose projections recover GR-type constrained Hamiltonian interiors, QM-type Hilbert/Schrodinger predictive interiors, and Bell/measurement-record boundary quotients. The task is not to quantize gravity, but to realize the constraint architecture from which gravitational and quantum structures are projected.

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